

30. Abstract Construction of Ideal Theory in Algebraic Number and Function Fields

Math. Ann. 96 (1927), pp. 26–61.

In what follows an abstract characterization is given of all those rings whose ideal theory agrees with the ideal theory of all integers of an algebraic number field: that is, whose ideals can be represented uniquely as products of powers of prime ideals. Among the best-known examples of such rings are also the ring of all integers of an algebraic function field in one indeterminate, and, more generally, in several indeterminates as soon as, following Kronecker, one restricts to ideals of highest dimension, i.e. passes to a certain quotient ring, the functional domain.

For the underlying commutative ring, the axioms completely equivalent to this ideal theory are the following:¹

- I. Divisor-chain condition.** Every chain of ideals in which each ideal is a proper divisor of the preceding one terminates after finitely many steps; in other words, in every divisor chain of ideals there is an index from which on all ideals are equal.
- II. Multiple-chain condition modulo every nonzero ideal.** Every chain of ideals — all of which are divisors of a fixed ideal different from the zero ideal — in which each ideal is a proper multiple of the preceding one terminates after finitely many steps.
- III.** Existence of the identity element for multiplication.
- IV.** Ring without zero divisors.
- V. Integral closedness in the quotient field.** Every element of the quotient field that is integral with respect to the ring belongs to the ring.

From these axioms I develop, step by step, an increasingly restricted ideal theory up to the desired one; conversely, I show that all these axioms are in fact satisfied there.

From the divisor-chain condition I it follows, as I showed in *Ideal Theory* §4, that an ideal is representable as the least common multiple of finitely many primary ideals belonging to distinct prime ideals. I recall this proof briefly here (§6), both because it can be simplified by using the concept of the ideal quotient, and because I want to correct an oversight: the original proof uses, at one point, the not-assumed existence of the identity element of multiplication.²

The assumption of the multiple-chain condition has the effect (§7) that, in the residue class ring modulo every ideal different from the zero ideal, a prime ideal cannot have a proper divisor, except for the unit ideal containing all elements. Consequently the primary components of these ideals are determined uniquely, except for those belonging to the unit ideal. In a somewhat less sharp form this result is already found in Masazo Sono;³ his assumption of the existence of a composition series in the residue class ring is equivalent to the “double chain condition” in this ring, as I show briefly at the end (§10). By double chain condition I mean the validity of both the divisor-chain and multiple-chain conditions without restriction to ideals different from the zero ideal.

If axiom III — existence of the identity element — is also satisfied, the unit ideal does not occur as an associated prime ideal. The primary components are therefore uniquely determined; they are pairwise coprime, and their least common multiple equals their product. Axioms I, II, III hold for all finite orders of an algebraic number field; already Dedekind (Dirichlet–Dedekind, *Zahlentheorie*, 3rd ed., 11th supplement,

¹For the definition of the basic concepts of ideal theory, compare for instance E. Noether, *Idealtheorie in Ringbereichen*, Math. Ann. 83 (1921), pp. 24–66 (cited below as *Ideal Theory*). Incidentally, the most essential concepts are also formulated briefly in the present paper, especially in §§1, 4, and 5. For completeness, some material is included that is not needed for the immediate purposes of this paper.

²P. Urysohn called my attention to this oversight; my own alteration of the proof was somewhat more cumbersome than the one given here, which is due to E. Artin. He had already filled the gap for himself earlier, and, independently of me, had also observed the simplification with the ideal quotient. — I take a further simplification, which avoids the introduction of the “reduced representation,” from a related question in W. Krull, *Algebraische Theorie der Ringe III*, Math. Ann. 92 (1924), pp. 181–213, Theorem I.

³Masazo Sono, *On the Reduction of Ideals*, Memoirs of the College of Science, Kyoto University, Series A, 7 (1924), pp. 191–204.

§172), without giving the proof in full detail, stated here the unique representation of an ideal as a product of pairwise coprime one-kind ideals (primary components).

From assumption IV — absence of zero divisors — it follows that the different powers of an ideal distinct from both the zero ideal and the unit ideal are all distinct, and that the quotient field exists. If finally axiom V of integral closedness in the quotient field is also satisfied, the primary ideals — uniquely determined because of IV — become powers of prime ideals, and the usual factorization theorem is obtained (§8). This latter proof rests on a consequence of integral closedness already drawn by Dedekind (3rd ed., §172) in the case of number fields: a genuinely fractional element can be represented as a quotient of integers in such a way that even the square of the numerator is not divisible by the denominator. In later presentations this consequence is replaced — likewise as a consequence of integral closedness — by the much less transparent generalized Gaussian theorem or by corresponding module theorems that say somewhat more; all of these are theorems whose application presupposes basis elements, whereas here one works directly with the ideals themselves.

In the converse direction (§9), the axioms distinct from axiom V follow almost directly; V follows from the proof that a genuinely fractional element can always be represented so that no power of the numerator is divisible by the denominator, which is therefore a strengthening of the consequence originally derived from V. This proof succeeds only after one draws from the ideal representation the usual conclusions: representation by principal ideals modulo a fixed ideal, and the theory of fractional ideals, i.e. of ideal quotients in the field. The fact that integral closedness follows from the representation by powers of prime ideals was also already known to Dedekind, as is shown by a remark in the 3rd edition, §172, p. 522, bottom.

Axioms I through V imply that the four decompositions which are generally distinct — into least pairwise coprime ideals, into mutually prime ideals, into largest primary components, and into irreducible ideals — coincide; conversely, from this coincidence and axioms I–IV, integral closedness follows. For rings with zero divisors, fulfillment of the remaining axioms — with V defined as integral closedness in the quotient ring — does not necessarily imply that the decompositions coincide, as the example of the residue class ring modulo certain ideals of a finite order shows (§9).

In §1 I sketch the theory of quantities integral with respect to a ring, up to Dedekind's consequence from integral closedness. In §2 I show how the chain conditions are transferred from the ring to modules consisting of linear forms, with this ring as coefficient and multiplier domain.⁴ From this I draw (§3) the conclusion that, on passing to the integral elements of an algebraic extension field of the first kind, axioms I to IV remain valid for every order, while for the order consisting of all integral elements, V is also valid. This fact — of course familiar for the algebraic number field — permits the subordination of the function fields mentioned at the beginning; in particular, by the simple proof that the functional domain derived from integral polynomials in several indeterminates satisfies the axioms, Kronecker's ideal theory is fully subsumed.⁵

The ideal theory subsequently developed does not presuppose these §§2 and 3, which serve only for classification. In §4 and §5 the foundations independent of axioms I through V are given; this brings out more sharply than in the original foundation which parts of the theory are independent of finiteness assumptions.

§1. Theory of Integral Quantities

Let there be given a commutative ring⁶ T without zero divisors, with identity element e for multiplication (axioms III and IV); in T a fixed subring R containing the identity element is distinguished. The concepts module, order, integral are throughout to refer to this fixed ring R , which later will be omitted from the

⁴This form of the module theorems is due to Prüfer (Neue Begründung der algebraischen Zahlentheorie, §2, Math. Ann. 94 (1924), pp. 198–243) and is more transparent than the customary transfer of the basis theorems.

⁵The difference appears only in the discriminant theorems, because the residue class ring of the functional domain modulo a prime number becomes an imperfect field, and hence extensions of the second kind can occur.

⁶As is known, a ring is defined once an equality relation is given in a set, and in addition two operations, addition and multiplication, are given satisfying the usual laws: addition is associative, commutative, and uniquely reversible; multiplication is associative — commutative when the ring is commutative — and distributive with respect to addition. If the underlying definition of equality is not set-theoretic identity, then, so that the same definition of equality is retained in every subsystem, such a subsystem (subring, module, ideal) must contain with any element all elements equal to it. Likewise, in every extension it must be assumed that the new definition of equality includes the old one. All concepts such as “finitely many elements,” “unique assignment,” etc. are meant in the sense of the equality definition; for example, “there is one and only one element” means that there is only one element up to equality. Incidentally, one can always pass from equality to set-theoretic identity by collecting equal elements into a class and introducing these classes as the new elements. The remarks given here on the definition of equality are due to R. Hölzer.

notation for brevity.⁷ Elements of R will be denoted by Latin letters, elements of T in general by Greek letters.

1. A system M of elements of T is called, as usual, an R -module — briefly, a module — if, together with two elements α and β , it also contains their difference, and with α also $r\alpha$, where r is any element of R . M is said to be divisible by N ,

$$M \equiv 0 \pmod{N},$$

and M is called a multiple, N a divisor, if every element of M is also contained in N . R -modules all of whose elements belong to R are called ideals in R .

Clearly the intersection — the least common multiple $[\dots, M_i, \dots]$ — of any number of modules from T is again a module; likewise T itself is a module. Thus the module derived from any system Σ of elements of T exists uniquely as the intersection of all modules containing Σ . The sum — the greatest common divisor $(\dots, M_i, \dots)$ — of any system of modules is the module derived from their union. The product MN of two modules is defined as the module derived from the elements $\mu\nu$, where μ runs through all elements of M and ν through all elements of N . The product therefore satisfies the associative and commutative laws, since these hold for the ring elements. The distributive law in T further gives, for modules, the distributive law

$$(\dots, M_i, \dots)N = (\dots, M_iN, \dots),$$

since by exactly this law in T both sides are the module derived from all elements $\mu_i\nu$.

Since R itself is a module, the condition that R is the multiplier domain can be expressed by $RM \equiv 0 \pmod{M}$. Since by hypothesis R contains the identity element, one also has $M \equiv 0 \pmod{RM}$, and hence $M = RM$.

A module M is called finite if there is at least one system Σ consisting of finitely many elements from which M is derived; the elements $\mu_1, \dots, \mu_n$ of Σ are called a module basis of

$$M = (\mu_1, \dots, \mu_n).$$

The sum and product of two — and hence of finitely many — finite modules are again finite, since they are derived from the union, respectively from the products, of the basis elements. The latter assertion follows from the representation $r_1\mu_1 + \dots + r_n\mu_n$ of all elements μ of M by the basis and from the commutative law of multiplication in T .

2. An extension ring of R lying in T — hence one that contains all elements of R — is called an R -order, briefly an order. The intersection of arbitrarily many orders in T is again an order, and likewise T is an order; hence the order derived from any system Σ of elements of T exists uniquely. Sums and products of orders are defined correspondingly as for modules; in particular $O^2 = O$ for every order.

Every order is at the same time an R -module; an order is called finite when it is a finite module. Since sums and products arise by forming module sums and products, sums and products of finite orders are again finite orders by 1.; further, the following transitive law holds: if A is a finite R -order and B is a finite A -order, then B is also a finite R -order. For B is at the same time an R -order, hence an R -module. If $\alpha_1, \dots, \alpha_m$ form a module basis of A with respect to R , and $\beta_1, \dots, \beta_n$ one of B with respect to A , then the products $\alpha_i\beta_j$ plainly form a module basis of B with respect to R .

3. An element α of T is called integral with respect to R , briefly integral, if the order R_α derived from α is finite — equivalently, if the divisor chain of modules

$$U_n = (\alpha^0, \alpha, \dots, \alpha^{n-1})$$

terminates after finitely many steps, $U_n = U_{n+1} = \dots$.

The definition also admits a second formulation, to be used in 4.: an element α of T is called integral if there is at least one principal module C different from the zero module — i.e. C is derived from an element $\gamma \neq 0$ — such that the product of the order R_α with C is a finite module; in other words, such that the chain of modules CU_n terminates after finitely many steps.⁸

⁷The theory of integral quantities remains valid if zero divisors are admitted, provided that in their place one assumes the divisor-chain condition in R , which by §2 also makes the proof of the lemma possible. Since this form of the theory of integral quantities is not used later, I only point to it in footnotes. All definitions under discussion remain valid in the presence of zero divisors; most of them require — as is known and easy to see — even weaker assumptions. What is not retained is Dedekind's formulation: "A number is integral when it has a hull." For with zero divisors present, the quotient of two finite modules need not remain finite; hence here the order is defined directly and not as a module quotient.

⁸If zero divisors are admitted in R , then the existence of a regular principal module must be required; that is, γ must be assumed to be a non-zero-divisor, a regular element, which in rings without zero divisors is simply $\gamma \neq 0$.

Finally the definition is also identical with the usual formulation: an element α of T is called integral if it satisfies at least one equation

$$\alpha^n + r_1\alpha^{n-1} + \cdots + r_n = 0,$$

where the r 's are elements of R .

These definitions are in fact identical. Since R_α agrees with the module derived from all powers of α , when R_α is finite one can always choose a module basis from these powers. If such a basis is given, for instance, by $e = \alpha^0, \alpha, \dots, \alpha^{n-1}$, this means that the chain of the U_n terminates at U_n ; conversely, $U_n = U_{n+1} = \cdots$ gives this basis. But the termination of the chain of the U_n is also identical with the existence of the equation

$$\alpha^n + r_1\alpha^{n-1} + \cdots + r_n = 0.$$

If R_α is finite, then CR_α is finite, so the chain CU_n terminates; conversely, from the finiteness of CR_α , hence from the termination of the CU_n , it follows that the U_n also terminate and hence that R_α is finite. Indeed, since C is assumed to be a principal module distinct from the zero module, $CU_n = CU_{n+1}$ also gives $U_n = U_{n+1}$.

The ring property of the integral quantities and the transitive law of integral dependence rest on the following

Lemma. Let O be a finite order in T , and let α be any element of O . Then the order R_α derived from α is finite; in other words, all elements of a finite order are integral.

The proof follows by the usual arguments. Let $\omega_1, \dots, \omega_n$ be a module basis of O . Since O is a ring, $\alpha\omega_i$ belongs to O . Thus

$$\alpha\omega_i = r_{i1}\omega_1 + \cdots + r_{in}\omega_n,$$

and hence

$$\det(r_{ij} - e_{ij}\alpha) = 0,$$

where $e_{ij} = 0$ for $i \neq j$ and $e_{ii} = e$; this proves that α is integral. For the vanishing of the determinant one uses the hypothesis that T , and hence O , has no zero divisors.⁹

The system G of all integral quantities in T forms an order. It contains R , for the elements of R are integral, since R is a finite R -module with the identity element as basis. But G also has the ring property: for the order $R_{\alpha,\beta}$ derived from two arbitrary integral quantities α, β is equal to the product $R_\alpha R_\beta$, and is therefore finite. Hence, by the lemma, $\alpha + \beta$ and $\alpha\beta$ are integral as elements of a finite order.

Transitive law of integral dependence. If O is an order consisting of integral quantities, then every element of T integral with respect to O is also integral with respect to R . By definition α satisfies an equation

$$\alpha^m + \omega_1\alpha^{m-1} + \cdots + \omega_m = 0,$$

and is therefore also integral with respect to the order $O' = R_{\omega_1, \dots, \omega_m}$ derived from $\omega_1, \dots, \omega_m$, which is finite as the product $R_{\omega_1} \cdots R_{\omega_m}$. By the transitive law given in 2., the finite O' -order O'_α derived from α is therefore also a finite R -order, and α is integral by the lemma as an element of a finite order.

4. The ring R is called integrally closed in T if every element of T integral with respect to R belongs to R . According to the second formulation of the definition of integral quantities in 3., an element α of T belongs to R if and only if there is at least one principal module C different from the zero module such that CR_α is a finite module.¹⁰

From the concept of an integrally closed ring Dedekind (3rd ed., §172) drew two important consequences, which make it possible to use this concept for ideal theory:

Dedekind consequence I. Let R be integrally closed in T , and assume further that the divisor-chain condition for ideals (axiom I) holds in R . If β is a nonintegral element of T , and $c \neq 0$ an element of R , then there is an exponent $\rho > 0$ such that in the sequence of elements

$$c, c\beta, \dots, c\beta^\nu, \dots$$

⁹If the divisor-chain condition is assumed in R but zero divisors are admitted, then the lemma is an immediate consequence of the module theorem in §2, according to which the chain condition transfers to the modules from O . This proof too is due, for the number field, to Dedekind (4th ed., §173, II) and is the first application of chain conditions in the literature. In the consequences to be given under 4., Dedekind instead uses the more special fact that the number of residue classes modulo an ideal in an algebraic number field is finite.

¹⁰If zero divisors are admitted in R , C must again be assumed to be a regular principal module; likewise the element c in Consequence I must be assumed regular.

the elements $c, \dots, c\beta^\rho$ are integral, and all the remaining ones are nonintegral.

If all elements $c\beta^\nu$ were integral, then by the integral closedness of R they would belong to R . Then the chain of modules $C, CB_1, \dots, CB_\nu, \dots$ — where C denotes the module derived from c , and B_ν the one derived from $1, \beta, \dots, \beta^\nu$ — would become a chain of ideals $C_\nu = (c, c\beta, \dots, c\beta^\nu)$, which terminates by hypothesis. But then, contrary to the assumption, R_β would be finite and β integral; hence some of the elements $c\beta^\nu$ are nonintegral. Furthermore, with any one nonintegral element all subsequent ones are nonintegral: for if $c\beta^r$ is integral, hence belongs to R , then for $\rho < r$ the element $c\beta^\rho$ satisfies

$$(c\beta^\rho)^r - c^{r-\rho}(c\beta^r)^\rho = 0$$

and is therefore integral. Thus if $c\beta^{\rho+1}$ is the first nonintegral element, then — since c is integral — $\rho > 0$, and this is the required exponent.

Specializing the extension ring T to the quotient field of R — that is, to the field obtained by adjoining all pairs of elements¹¹ — gives the following:

Dedekind consequence II. Let R be integrally closed in its quotient field K , and let the divisor-chain condition for ideals hold in R . If β is a nonintegral element of K , then β admits a representation as a quotient of elements of R ,

$$\beta = m/n,$$

such that m^2/n is also nonintegral.

Put $\beta = b/c$ with $c \neq 0$. Then in the sequence $c, c\beta = b, c\beta^2, \dots$ the first two elements are certainly integral, so $\rho > 1$, where ρ is the exponent of Consequence I. If one sets

$$m = c\beta^\rho, \quad n = c\beta^{\rho-1},$$

then $m/n = \beta$, and $m^2/n = c\beta^{\rho+1}$ is nonintegral.

The transitive law of integral closedness has the following form: if R is any ring in T and G denotes the system of all quantities in T integral with respect to R , then G also consists of all quantities of T integral with respect to G . For every quantity integral with respect to G is also integral with respect to R , by the transitive law in 3., and hence belongs to G .

§2. Chain Conditions in Finite Module Domains

The module theorem by which the chain conditions are transferred occurs, in the application given here, only for modules of an extension ring. Since the proof is the same in the case of a general module domain, such a domain will be taken as the starting point.

A system M of elements $\alpha, \beta, \dots$ is called a module domain with respect to a ring R if two operations are given in M , each leading uniquely to elements of M : an additive combination of the elements and a multiplication of the elements by elements of R ; if M is an Abelian group with respect to addition, while multiplication satisfies the associative law; and if the distributive law holds in both forms.¹²

For a module domain, all module definitions given in §1, 1. that do not refer to multiplication plainly remain valid. In particular, M is called a finite module domain if there are finitely many elements $\xi_1, \dots, \xi_k$ of M such that M equals the module derived from $\xi_1, \dots, \xi_k$, hence equals the system of all linear forms $r_1\xi_1 + \dots + r_k\xi_k$ when R has an identity element; otherwise additional terms $n_i\xi_i$ occur.

Module theorem. Let M be a finite module domain with respect to a commutative ring R with identity element, and suppose that in R the divisor- respectively multiple-chain condition for ideals holds. Then in M the divisor- respectively multiple-chain condition for modules holds.

The proof rests on assigning uniquely to every module in M a system of finitely many ideals from R in such a way that to a proper divisor respectively proper multiple of the module there corresponds each time at least one proper divisor respectively multiple ideal.

An element of M is said to have length i if it can be written in at least one way as a linear form in $\xi_1, \dots, \xi_i$, while it has no representation as a linear form in $\xi_1, \dots, \xi_{i-1}$. To an arbitrary module W in M assign now k ideals $\mathfrak{a}_1, \dots, \mathfrak{a}_k$ from R : let W_i mean the module of all elements of W of length $\leq i$, and let $\mathfrak{a}_i$ denote the system of all coefficients of ξ_i in W_i . First one obtains:

¹¹In Steinitz's familiar formulation, J. f. M. 137. If zero divisors are admitted in R , then the quotient ring must replace the quotient field, by adjoining all quotient pairs whose denominator is a regular element of R . Here Dedekind consequence II is no longer valid, for n will not in general be a regular element.

¹²Compare *Ideal Theory*, §9.

If B is a proper divisor of W , and if $\mathfrak{b}_1, \dots, \mathfrak{b}_k$ are the ideals assigned to B , then among the $\mathfrak{b}_i$ there is at least one proper divisor of the corresponding $\mathfrak{a}_i$. For from $B \equiv 0 \pmod{W}$ it follows that $B_i \equiv 0 \pmod{W_i}$, and hence $\mathfrak{b}_i \equiv 0 \pmod{\mathfrak{a}_i}$ for $i = 1, 2, \dots, k$. By hypothesis there must be elements in B not contained in W , and hence such elements of smallest length ℓ . Then $\mathfrak{b}_\ell$ is a proper divisor of $\mathfrak{a}_\ell$; indeed $\mathfrak{b}_\ell \equiv 0 \pmod{\mathfrak{a}_\ell}$ when

$$\beta = b_1\xi_1 + \dots + b_\ell\xi_\ell$$

is such an element of smallest length in B . If $\mathfrak{b}_\ell = \mathfrak{a}_\ell$, there exists an element

$$\alpha = a_\ell\xi_\ell + \dots + a_1\xi_1 \equiv 0 \pmod{W},$$

and therefore an element $\beta - \alpha \in B$, $\beta - \alpha \notin W$, of smaller length than ℓ .

Now let a divisor or multiple chain

$$W^{(1)}, W^{(2)}, \dots, W^{(\nu)}, \dots$$

be given; suppose the corresponding chain condition holds in R . If $\mathfrak{a}_{1\nu}, \dots, \mathfrak{a}_{k\nu}$ are the ideals assigned to $W^{(\nu)}$, then the sequences

$$\mathfrak{a}_{i1}, \mathfrak{a}_{i2}, \dots$$

are divisor respectively multiple chains for $i = 1, \dots, k$. Hence by hypothesis there is an index ν_0 such that the ideal $\mathfrak{a}_{i\nu_0}$ is equal to all subsequent ideals for every i . But then, by what was just proved, the module $W^{(\nu_0)}$ is equal to all subsequent modules; thus the chain conditions are transferred.

This immediately gives the following:

Consequence of the module theorem. If in R the multiple-chain condition holds modulo every ideal different from the zero ideal, then in M the multiple-chain condition holds modulo every module C whose assigned ideals $\mathfrak{c}_1, \dots, \mathfrak{c}_k$ are all different from the zero ideal. Under these hypotheses the multiple chains $\mathfrak{a}_{i1}, \dots, \mathfrak{a}_{i\nu}, \dots$ terminate after finitely many steps.

Additional remark. If R is a ring without identity element, then the divisor-chain condition and the multiple-chain condition modulo ideals different from the zero ideal are still transferred. Namely, instead of M one considers the system M' of all elements of the form

$$a_1\xi_1 + \dots + a_k\xi_k + n_1\xi_1 + \dots + n_k\xi_k,$$

which again passes into M when the additional integral multiples $n_i\xi_i$ are replaced by elements of M . To this M' one can assign, correspondingly as above, $2k$ ideals, where the $\mathfrak{a}_i$ are ideals from R and the $\mathfrak{n}_i$ are ideals of the rational integers. Since for the $\mathfrak{n}_i$ both the divisor-chain condition and the multiple-chain condition modulo every nonzero ideal are satisfied, the proof for the divisor-chain condition remains valid for M' and hence for M ; for the multiple-chain condition, however, one must require that the $2k$ ideals assigned to C respectively C' are all different from the zero ideal.